

segregation: Segregation Analysis, Inference, and Decomposition in Python

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

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Submitted: 01 January 1970

Published: unpublished

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Summary

segregation is an open-source Python library for the measurement, analysis, inference, and decomposition of social and spatial segregation patterns. Developed as part of the PySAL ecosystem, the package provides a comprehensive framework for quantifying segregation across a wide range of demographic, socioeconomic, and geographic contexts. It supports both traditional aspatial measures and advanced spatially explicit approaches, enabling researchers to investigate how population groups are distributed across space and how these patterns evolve over time.

Segregation analysis is a central topic in urban studies, geography, sociology, demography, and public policy. The segregation package implements a large collection of segregation indices, including single-group, multigroup, local, and spatial measures. These methods capture different dimensions of segregation such as evenness, exposure, concentration, clustering, centralization, and spatial proximity. The package accommodates a variety of spatial representations, including adjacency relationships, distance-based interactions, and network-based structures, allowing users to select measures that align with the theoretical and empirical characteristics of their study area.

Beyond point estimation, segregation provides a robust inferential framework for evaluating the statistical significance of segregation measures. Users can perform single-value and comparative hypothesis tests using simulation-based approaches, facilitating rigorous assessments of whether observed segregation patterns differ from those expected under specified null models. The package also includes decomposition methods that separate differences in segregation levels into components attributable to demographic composition and spatial structure, offering deeper insight into the processes underlying segregation dynamics.

Designed to operate natively with Pandas and GeoPandas data structures, segregation integrates seamlessly into contemporary Python-based spatial data science libraries and therefore minimizes friction in typical workflows. As a component of the Python Spatial Analysis Library (PySAL) ecosystem (S. J. Rey et al., 2022; S. J. Rey & Anselin, 2007), it follows shared principles of interoperability, transparency, reproducibility, and methodological rigor. By providing accessible, well-documented, and extensible implementations of state-of-the-art segregation measures, segregation serves both applied researchers investigating social inequality and methodological developers advancing quantitative approaches to segregation analysis.

Statement of need

Residential segregation, inequality, and spatial separation of population groups remain central concerns in urban studies, sociology, demography, geography, and public policy.

41 Researchers and practitioners frequently seek to quantify the extent to which social groups are
42 unevenly distributed across neighborhoods, cities, and regions, as well as to understand how
43 segregation patterns vary across space and time. The increasing availability of demographic
44 and socioeconomic data has expanded opportunities for segregation analysis, but it has also
45 highlighted the need for accessible, transparent, and reproducible computational tools capable
46 of implementing a wide range of segregation measures and inferential procedures.

47 Traditional software environments for segregation analysis often provide only a limited set of
48 indices, rely on proprietary platforms, or require researchers to implement methods manually.
49 These limitations can hinder reproducibility, reduce methodological transparency, and create
50 barriers for comparing results across studies. Furthermore, advances in segregation research have
51 introduced numerous spatially explicit measures, local indicators, decomposition techniques,
52 and simulation-based inference methods that are not consistently available within existing
53 analytical software.

54 Within the Python ecosystem, support for segregation analysis was historically fragmented.
55 While libraries such as Pandas and GeoPandas provide essential data structures for handling
56 tabular and spatial data, they do not natively implement segregation measures or associated
57 inferential frameworks. Consequently, researchers often relied on custom scripts, isolated
58 software implementations, or statistical packages in other programming languages, resulting in
59 inconsistent workflows and difficulties in reproducing analyses.

60 segregation addresses these challenges by providing:

- 61 ■ A comprehensive and well-documented API for segregation measurement and analysis
- 62 ■ Native integration with Pandas, GeoPandas, and the broader PySAL ecosystem
- 63 ■ Support for a large collection of aspatial, spatial, local, and multigroup segregation
64 indices
- 65 ■ Functions for measuring and visualizing multiscalar segregation profiles
- 66 ■ Simulation-based inference procedures for evaluating the statistical significance of
67 segregation measures
- 68 ■ Decomposition methods that facilitate the investigation of demographic and spatial
69 sources of segregation
- 70 ■ A strong emphasis on transparency, reproducibility, and methodological extensibility

71 These capabilities make segregation particularly valuable for researchers and practitioners in
72 fields such as urban studies, sociology, demography, geography, public policy, public health,
73 and regional science, where understanding patterns of spatial inequality and social separation
74 is a fundamental analytical objective.

75 State of the field

76 segregation is a component of the PySAL ecosystem, which provides a comprehensive suite
77 of tools for spatial analysis in Python. Within this ecosystem, the packages are divided into
78 four types (example libraries in parentheses):

- 79 ■ **Lib**: Core spatial data structures, file IO. Construction and interactive editing of spatial
80 weights matrices & graphs. Alpha shapes, spatial indices, and spatial-topological
81 relationships (`libpysal`)
- 82 ■ **Explore**: Modules to conduct exploratory analysis of spatial and spatio-temporal data
83 (`esda`, `giddy`, `inequality`, etc.)
- 84 ■ **Model**: Estimation of spatial relationships in data with a variety of linear, generalized-
85 linear, generalized-additive, and nonlinear models (`spreg`, `spopt`, `tobler`, `spglm`, etc.)
- 86 ■ **Viz**: Visualize patterns in spatial data to detect clusters, outliers, and hot-spots
87 (`mapclassify`, `splot`, etc.)

88 segregation is present in the **Explore** set of libraries and addresses the specific problem of
89 residential segregation.

90 Compared to desktop GIS platforms, segregation offers several advantages:

- 91 ▪ **Reproducibility:** Workflows can be scripted and version-controlled
- 92 ▪ **Transparency:** Methods and assumptions are explicit and inspectable
- 93 ▪ **Extensibility:** Users can modify or extend algorithms for research purposes
- 94 ▪ **Integration:** Segregation analysis can be embedded within larger data science pipelines,
- 95 including machine learning and statistical modeling

96 Segregation can also be assessed with other software, such as the R packages *OasisR* (Tivadar,
97 2019), *seg* (Hong, 2011) or *segregation* (Elbers, 2021). However, PySAL's segregation
98 module distinguishes itself in several critical ways: supports over 40 distinct segregation indices
99 out-of-the-box, it allows simultaneous computation of multiple indices across multi-scalar
100 geographical frameworks with fewer software dependencies, it integrates spatial networks using
101 topological relationships allowing users to measure social separation based on real-world street
102 networks rather than assuming straight-line distances (Euclidean), it also has a set of function
103 for evaluating statistical significance using simulations, and it also allows users to decompose
104 segregation scores—identifying exactly how much segregation occurs due to spatial context.

105 In addition, segregation provides a native solution for Python users, aligning with the growing
106 adoption of Python in geospatial and data science communities.

107 Software design

108 segregation is designed with attention to flexibility, usability, and interoperability between
109 its core functions. The library is organized into different subparts such as *singlegroup*,
110 *multigroup*, *local*, *inference*, *decomposition*, *batch*, *network*, and *dynamics*.¹ Segregation
111 measures are built using Python classes which can be integrated in subsequent steps, such as
112 inference or decomposition. The module is structured toward two kinds of segregation indices:
113 'spatially-explicit' and 'spatially-implicit'. The former includes space as part of its original
114 formula. The latter uses the Sean F. Reardon & O'Sullivan (2004) approach to state that any
115 segregation index is a spatial index if you transform the data properly.

116 Additionally, segregation is developed with testing and documentation standards consistent
117 with the Scientific Python ecosystem, ensuring reliability and maintainability.

118 Core Functionality

119 segregation organizes its functionality around the type of segregation analysis the user is
120 interested in, and each subpart is explained as follows.

121 Single and Multigroup Indices

122 Single-group measures assess segregation between two different groups in a given location (i.e.,
123 one group vs. everyone else). Multigroup segregation evaluates the simultaneous separation of
124 all groups in a population (e.g., the distribution of White, Black, Asian, and Hispanic residents)
125 across areas.

126 Currently, segregation has over 40 indices available which represents, to our knowledge, the
127 broadest range of indices available for a user in any software. Also, the user can fit many
128 indices at once with a wrapper function in the *batch* module.

129 Local Indices

130 Unlike global indices that summarize an entire metropolitan area into a single value, local indices
131 decompose segregation to the individual geographic unit level. Using these disaggregated

¹Cortes et al. (2020) introduced segregation but many new implementations were developed recently and the API underwent a major revision.

measures helps identify precise spatial clusters where social isolation is most acute, uncovering micro-level dynamics that global metrics often mask. Currently, segregation has seven local indexes.

Multiscalar

The multiscalar profile (Sean F. Reardon et al., 2008) is a tool for measuring spatial segregation dynamics—the way that a segregation index changes values as the concept of a neighborhood changes, and what that tells us about macro versus micro patterns of segregation. The core idea is to calculate a segregation statistic, then expand the spatial scope of a neighborhood, recalculate the statistic, and repeat.

The package has a wrapper named `compute_multiscalar_profile` which can be used in a workflow to build these profiles.

Simulation based Inference

PySAL's segregation module also addresses whether segregation index values are statistically significant under different specifications of a null hypothesis using Monte Carlo simulations. Currently, for single value inference the module can generate bootstrap replications of the units with replacement (`bootstrap`), multinomial with restricted conditional probabilities (`systematic`), binomial with fixed parameters (`evenness`), geographic unit-level randomization (`geographic_permutation`), among others. For two-value inference, the user can specify resampling to generate distributions of the segregation index for each index (`bootstrap`), generate a synthetic dataset for each region through counterfactual estimates (`composition`), among others.

The correct specification of a null hypothesis is a crucial part of this framework as different null hypotheses can lead to markedly different inferences, and also different segregation indexes, which can assess different segregation dimensions (i.e. `evenness`, `exposure`, `concentration`, `centralization`, and `clustering`) may not be appropriate for some specifications, especially for comparative inference.

Decomposition

The decomposition approach implemented in the PySAL segregation module provides a framework for comparative segregation analysis that disentangles observed differences in segregation levels into two fundamental components: population composition and spatial structure. Building on the framework proposed by S. Rey et al. (2021), the method uses spatially explicit counterfactual distributions combined with a Shapley value decomposition to determine how much of the difference between two segregation measures (whether across cities, time periods, or spatiotemporal contexts) is attributable to differences in demographic composition versus differences in the spatial arrangement of populations and areal units.

The method can be applied to a variety of segregation indices and comparison settings, including cross-sectional, temporal, and spatiotemporal analyses. By moving beyond simple comparisons of index values, the decomposition provides a richer interpretation of segregation dynamics and helps identify whether differences arise primarily from changes in population composition or from the spatial organization of neighborhoods. In the module, this functionality is available in the `DecomposeSegregation` class of the `decomposition` subpart.

Example workflow

Assuming you have a dataset (pandas or geopandas DataFrames) named `data_1`, a typical workflow in segregation can be implemented as follows:

```
from segregation.singlegroup import Dissim
```

```
seg_index_1 = Dissim(
    data_1,
    group_pop_var='group_A',
    total_pop_var='total_population'
)
```

176 This snippet estimates the Dissimilarity index of a specific sub-population (group with
177 characteristic A) of your dataframe and can be accessed through the statistic attribute.
178 If the user is interested in assessing statistical significance of the index, it can simply be
179 implemented as:

```
from segregation.inference import SingleValueTest
```

```
inference_result = SingleValueTest(
    seg_index_1,
    null_approach='bootstrap'
)
```

180 This approach assumes the bootstrap approach for the generation of the Monte Carlo iterations,
181 and the user can access the pseudo p-value estimated from the simulations through the p_value
182 attribute.

183 To compute all single group indices in one go, the package provides a wrapper function in the
184 batch module:

```
from segregation.batch import batch_compute_singlegroup
```

```
all_singlegroup = batch_compute_singlegroup(
    data_1,
    group_pop_var='group_A',
    total_pop_var='total_population'
)
```

185 To compute multiscalar profiles of, for example, a Gini index, the user can rely on the dynamics
186 module and specify:

```
from segregation.dynamics import compute_multiscalar_profile
```

```
gini_profile = compute_multiscalar_profile(
    data_1,
    segregation_index=Gini,
    group_pop_var='group_A',
    total_pop_var='total_population',
    distances=range(500,5500,500)
)
```

187 In terms of comparative segregation indexes, it is possible to assess statistical significance
188 between two measures using the TwoValueTest of the inference module as well as decompose
189 the comparison using DecomposeSegregation from decomposition. Assume the user would
190 like to compare the segregation of group_A between data_1 and another spatial context data_2.
191 Therefore, the code below depicts both analysis.

```
from segregation.inference import TwoValueTest
from segregation.decomposition import DecomposeSegregation
```

```
seg_index_2 = Dissim(
    data_2,
    group_pop_var='group_A',
    total_pop_var='total_population'
```

```
)

two_value_result = TwoValueTest(
    seg_index_1,
    seg_index_2,
    null_approach='bootstrap'
)

decomposition_result = DecomposeSegregation(seg_index_1, seg_index_2)
```

192 **Figure 1** depicts the black + brown share in total population in each census tract of 2022 in
 193 two major Brazilian cities: Belo Horizonte and Porto Alegre. Clear visual synthesis reveals a
 194 consistent pattern of socio-spatial segregation, characterized by a “center-periphery” racial
 195 gradient in both cities.

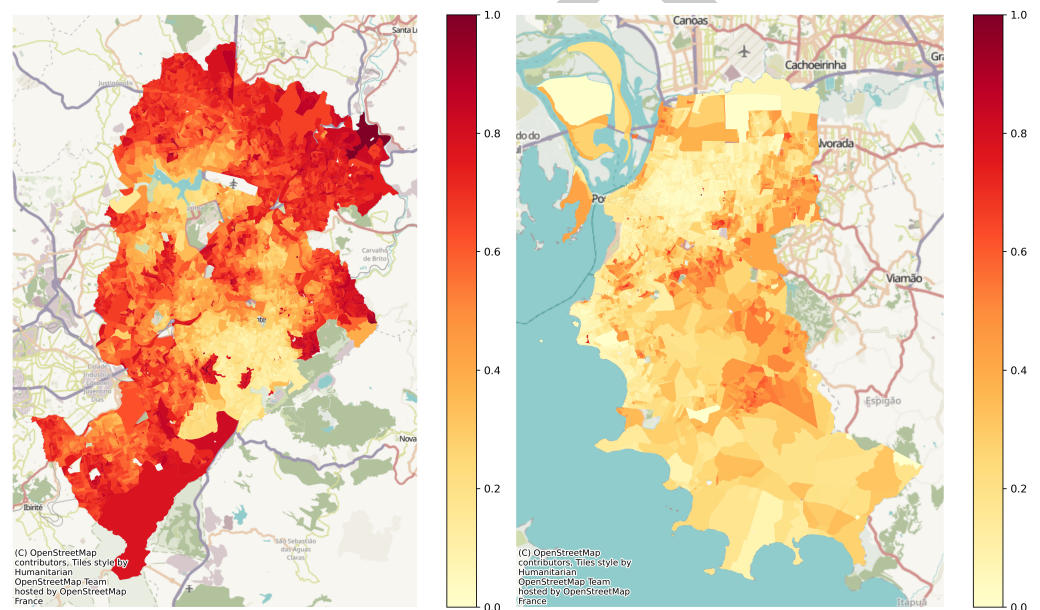


Figure 1: Belo Horizonte (left) and Porto Alegre (right) black + brown share in total population in each census tract of 2022.

196 **Figure 2** presents an example of comparative analysis of selected segregation indices via Monte
 197 Carlo bootstrap simulations using the PySAL segregation module that reveals significant
 198 structural differences in the racial landscapes of Porto Alegre and Belo Horizonte. For
 199 global indices such as Dissimilarity (Dissim), Spatial Dissimilarity, and the Gini index, the
 200 non-overlapping distributions indicate that Porto Alegre exhibits statistically higher levels
 201 of unevenness in population distribution compared to Belo Horizonte. Conversely, the
 202 Isolation and Distance Decay Isolation metrics show a dramatic reversal; Belo Horizonte
 203 (red) displays significantly higher isolation values (concentrated above 0.60) than Porto Alegre
 204 (blue, approximately 0.37), suggesting that despite lower overall dissimilarity, Black and Brown
 205 residents in Belo Horizonte are much more likely to live in tracts with high intra-group exposure.
 206 Furthermore, the stark divergence in Relative Clustering, where Porto Alegre's distribution
 207 is shifted significantly to the right suggests a more intense spatial aggregation of minority
 208 groups into contiguous clusters in the southern capital. In contrast, measures like Entropy
 209 and Relative Concentration show considerable distributional overlap, indicating that for these
 210 specific dimensions of segregation, the differences between the two cities may not reach
 211 statistical significance.

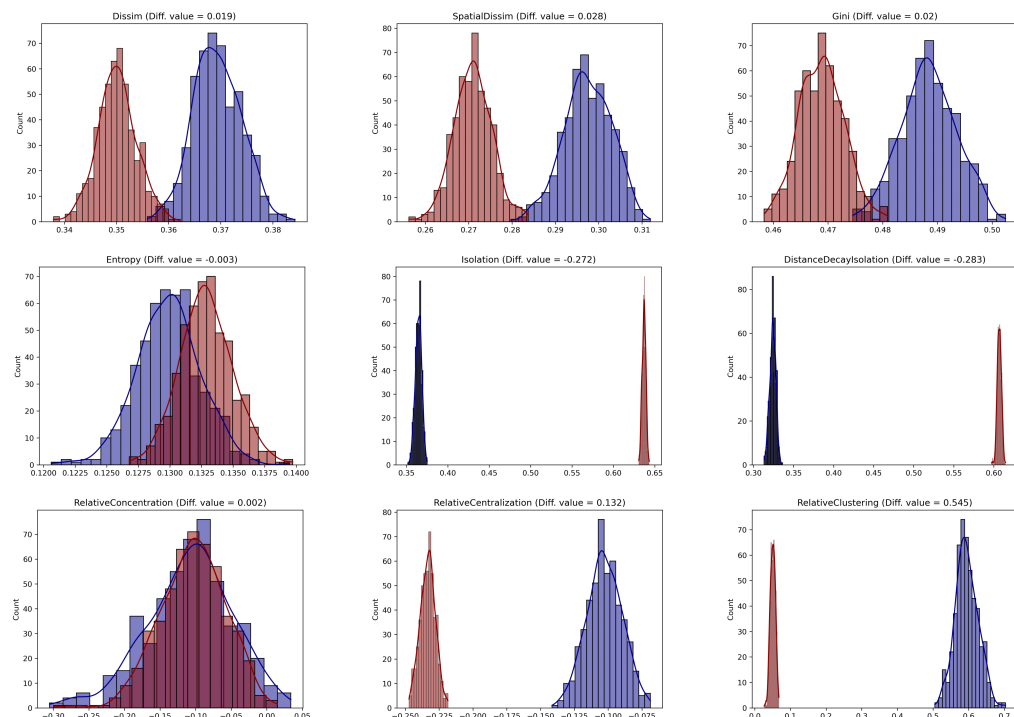


Figure 2: Example of segregation usage for comparative 2022 census tracts: Porto Alegre (blue) vs. Belo Horizonte (red) distributions under bootstrap simulation

Research impact statement

The package is actively used by the research community to assess segregation in many different contexts. Cortes et al. (2020) compared Los Angeles and New York racial segregation structures. Knaap & Rey (2024) uses it to compute the spatial information theory index ($\hat{H}$) for 380 U.S. metropolitan areas, S. J. Rey et al. (2024) relies on PySAL's segregation to measure multigroup dissimilarity in the school-neighborhood nexus, and Wei et al. (2022) employs its dissimilarity index within an optimization model to minimize racial segregation in school districts. The module also underpins comparative segregation analytics (S. Rey et al., 2021) and analyses of historical redlining legacies (S. Rey & Knaap, 2022), demonstrating its role as a core computational engine across urban science, education policy, and spatial demography.

In spatial data science education, segregation has become part of many curricula. It is included in pedagogical resources including textbooks (Knaap, 2026), and is often taught in global conferences like SciPy.²

AI usage disclosure

No generative AI or LLMs were used for code development in segregation; however, they were used for grammar and spelling corrections during the writing of this paper.

²<https://www.youtube.com/watch?v=4AHJVMs7iH4>

Acknowledgements

segregation is developed as part of the PySAL community, which brings together researchers and developers working on spatial analysis methods and software. The project builds on decades of research in segregation, urban, and spatial data science, and benefits from contributions across the open-source geospatial community.

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